Direction-Aware Graph Representation for Deep Reinforcement Learning-Based Droplet Routing on MEDA Biochips

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Abstract

Micro-electrode-dot-array (MEDA) biochips move nanolitre droplets across a grid of micro-electrode cells (MCs) whose electrodes degrade with use. Deep reinforcement learning has been applied to route droplets on such chips by encoding the chip state as an image with a convolutional neural network (CNN) and training the policy with proximal policy optimization (PPO). This paper represents the chip instead as a graph with one node per MC and eight-neighbour edges, and encodes it with a direction-aware graph convolutional network (GCN) that applies a separate weight matrix to each neighbour direction, followed by global max pooling. We show that isotropic message passing is insufficient, because it yields an embedding that is invariant to the mirror and rotation symmetries of the chip, which motivates direction-specific weights, and that the direction-aware encoder is equivalent to a fully convolutional network with global max pooling. With the simulator, reward and PPO learner unchanged, the direction-aware GCN, with 76 k encoder parameters against 29.7 M for the CNN, converged in all five training seeds on 30×30 chips after 134 ± 24 k environment steps, whereas the CNN converged only after 819 k steps in a single 100-epoch run (one seed). With a budget of 40 epochs, it routed on average at least 99.6% of held-out jobs (every run at least 99.2%) on chips of up to 100×100 MCs, while the success rate of the CNN fell from 87.4% to 15.6%; on 120×120 chips it reached 94.2% against 10.2%. On chips with 10% and 20% faulty MCs it routed 94.0% and 82.2% of the jobs against 80.4% and 69.3%. Its convergence within 6–10 epochs on 30×30 chips matches the training speed reported for the original CNN, which neither the released implementation nor our implementation of that CNN attains.

Index Terms

Digital microfluidics, MEDA biochips, droplet routing, deep reinforcement learning, graph convolutional networks.

I. Introduction

Digital microfluidic biochips automate biochemical protocols by manipulating discrete droplets on an array of electrodes through electrowetting-on-dielectric [1], [2]. The micro-electrode-dot-array (MEDA) architecture divides the array into thousands of identical micro-electrode cells (MCs), each with its own actuation and sensing circuitry [3], [4]. A droplet covers a rectangle of MCs and is moved by actuating the MCs adjacent to its leading edge. Repeated actuation degrades an MC, so that the force it exerts decreases and droplet movement becomes stochastic [5]. A router must therefore choose, at every control cycle, a direction that brings the droplet to its goal quickly while accounting for the sensed health of the MCs.

Elfar et al. [5], [6] formulated this task as a reinforcement-learning problem: a convolutional neural network (CNN) encodes the MC health, droplet and goal as a three-channel image, and proximal policy optimization (PPO) [7] learns the routing policy. The CNN treats the chip as an image of fixed size: chips of any size are resampled to 30×30 pixels, and 29.5 of the 29.7 million parameters of the network lie in a fully connected layer whose size is tied to that resolution.

A MEDA chip, however, is physically a set of cells with fixed neighbourhoods, which is naturally described as a graph. This paper investigates a graph representation of the chip state as the input of the PPO learner. Its contributions are as follows.

- A graph formulation of MEDA droplet routing with one node per MC and eight-neighbour edges, encoded by a direction-aware graph convolutional network (GCN) with global max pooling whose 76 k parameters do not depend on the chip size.

- An analysis showing that isotropic message passing cannot distinguish a routing job from its mirror images and rotations, and that the direction-aware encoder is equivalent to a fully convolutional network with global max pooling; both statements are verified by automated tests.
- A controlled comparison with the CNN–PPO method of [5] on chips of 30×30 to 120×120 MCs and on chips with faulty MCs, with up to five training seeds, together with a training comparison of our implementation of the baseline against the authors' original implementation [8].

II. Related Work

Droplet routing. Early methods routed droplets on ideal electrode grids as part of biochip synthesis [9]. Elfar et al. [10], [11] modelled MC degradation explicitly and synthesized adaptive routing strategies with stochastic games, which are reliable but costly for large chips. Liang et al. applied deep reinforcement learning to droplet routing on digital microfluidic biochips [12] and to the parallel control of multiple droplets on MEDA biochips [13]. Elfar et al. [5], [6] route droplets on MEDA biochips with a CNN–PPO agent that observes the sensed MC health, uses an action space whose step size adapts to the droplet size, and transfers learning across chip sizes.

Graph neural networks and reinforcement learning. Message-passing networks [14] update node embeddings from their neighbours; the GCN of Kipf and Welling [15] uses a normalized neighbourhood average, and relational GCNs [16] learn a separate weight matrix per edge type. Xu et al. [17] analysed how aggregation and readout functions limit the discriminative power of message-passing networks, and Beaini et al. [18] argued that isotropic aggregation cannot capture directions. In reinforcement learning, relational inductive biases have been introduced into deep agents [19], grid observations have been represented as graphs whose edge types encode spatial relations [20], and an edge-based graph network encodes the netlist for chip macro placement with PPO [21]. To our knowledge, no prior work represents the MEDA electrode array as a graph for a reinforcement-learning router.

III. Problem Formulation

Chip model. A $W\times H$ chip holds MCs indexed by (x,y). A routing job moves a droplet, a rectangle of MCs, from a start to a goal location within a routing zone, defined as the bounding box of start and goal enlarged by three MCs. Following [5], the degradation of MC (i,j) after n_{ij} actuations and its sensed health are

$$D_{ij}=\tau_{ij}^{\,n_{ij}/c_{ij}},\qquad H_{ij}=\min\!\big(\lfloor 2^{b}D_{ij}\rfloor,\ 2^{b}-1\big), \tag{1}$$

with $\tau_{ij}\in[0.5,0.7]$, $c_{ij}\in[500,800]$ and a $b=2$-bit health sensor. A move succeeds with a probability determined by the degradation of the MCs at the leading edge.

Markov decision process. The state is a three-channel $W\times H$ grid holding the MC health inside the routing zone ($H/2^{b}$, and 0 outside), the droplet and the goal. The action is one of eight directions; the step size adapts to the droplet size and never overshoots the goal (Algorithm 1 of [5]). The reward, with the coefficients of the reference implementation [8], is $+0.5\,\Delta d$ for a move that reduces the distance d to the goal, $0.8\,\Delta d-1$ otherwise, -1 for an action that would leave the routing zone, and $+100$ on arrival. An episode ends at the goal or after $k_{\max}=W_h+H_h$ control cycles, where $W_h\times H_h$ is the size of the routing zone.

Baseline. The CNN of [5] (Table I there) has three 3×3 convolutional layers with 64, 128 and 128 filters (stride 1, zero padding, as in [8]) and a 256-unit fully connected layer; linear actor and critic heads share it. Every chip is resampled to a 30×30 observation by area averaging.

IV. Direction-Aware Graph Representation

Graph. Each MC is a node with index $y\,W+x$ and features $X_i=[\text{health}_i,\text{droplet}_i,\text{goal}_i]$, copied from the observation without renormalization. Edges connect every MC to its eight spatial neighbours in both directions,

$$|E|=2\big[(W-1)H+W(H-1)+2(W-1)(H-1)\big], \tag{2}$$

i.e. 6 844 directed edges on a 30×30 chip. The graph always has the native resolution of the chip.

Direction-aware message passing. Each of $L=3$ layers of the direction-aware GCN (DA-GCN) updates node i as

$$h_i^{(l+1)}=\mathrm{ReLU}\Big(W_0^{(l)}h_i^{(l)}+\sum_{r=1}^{8}\sum_{j\in\mathcal{N}_r(i)}W_r^{(l)}h_j^{(l)}+b^{(l)}\Big), \tag{3}$$

where $\mathcal{N}_r(i)$ contains the neighbour of MC i in direction r, if it exists, and $h_i^{(0)} = X_i$. Directions are derived from the node indices, so the graph carries no edge attributes. With hidden dimension $d = 64$, the encoder has 75 648 parameters for any chip size. A global max pooling readout, $g_k = \max_i h_{i,k}^{(L)}$, gives a fixed-size embedding $g \in \mathbb{R}^{64}$, from which linear heads compute the action probabilities and the state value as in the baseline. The simulator, reward, action space and PPO settings are those of the baseline, so the encoder is the only difference between the two methods.

Why isotropic message passing is insufficient. Every mirror and rotation of a square chip (the dihedral group D_4) maps the eight-neighbour grid onto itself. An isotropic GCN [15], which weights all neighbours alike, therefore produces node embeddings of a transformed job that are a permutation of those of the original job, and any readout that ignores node order assigns both the same graph embedding. A policy computed from this embedding cannot distinguish a job from its mirror images and rotations: for a goal that lies straight along one axis from the droplet, the eight transformed jobs require each of the four axis directions twice, so such a policy selects the direction of the goal with probability at most $1/4$ on average over them.

Equivalence with convolution. Each direction-aware layer is a 3×3 convolution with stride 1 and zero padding: $W_0^{(l)}$ is the centre tap, $W_r^{(l)}$ is the tap at the position of the neighbour in direction r, and a neighbour that does not exist at the chip border contributes nothing, as zero padding does. The encoder is therefore a three-layer fully convolutional network followed by global max pooling, whose convolutional layers are translation-equivariant and whose size is independent of the chip size; a similar correspondence was noted by Jiang et al. [20]. Both this equivalence and the invariance of the isotropic GCN are verified numerically by automated tests. Each node embedding depends on the 7×7 block of MCs within three hops.

V. Experimental Setup

The simulator is implemented as a Gymnasium environment [22], and both methods are trained with the PPO implementation of Stable-Baselines3 [23]; Table I lists the settings. Training proceeds in epochs of 2^{14} environment steps. After every epoch the deterministic policy is evaluated on a fixed set of 500 jobs; this evaluation determines the learning-rate decay and the convergence point, defined as the first of three consecutive evaluations with at least 95% success. After training, each final model is evaluated on 500 held-out jobs drawn with a separate random seed, identical for both methods; these held-out results are reported unless stated otherwise. Results over several seeds are given as mean $\pm$ sample standard deviation, and differences are tested with Welch's t-test across seeds or, for single runs, with an exact McNemar test on the paired jobs. The CNN runs at 100×100 and 120×120 and the runs with seed 0 at 30×30 to 60×60 (40 epochs) were trained on Tesla T4 GPUs; all other runs used an A100 server. The DA-GCN runs at 100×100 and 120×120 were repeated there for the full 40 epochs because the T4 runs had reached a session time limit after 34 and 21 epochs.

TABLE I
Experimental Settings

Item	Value
Chips	30×30 to 120×120 MCs; healthy, or 10%/20% fully degraded MCs in 2×2 clusters redrawn every episode
Droplets	9 sizes, $w, h \in \{2, \ldots, 6\}$, $w/h \in [0.8, 1.25]$
Observation	CNN: 30×30 (resampled); GCN: native grid
PPO	8 environments, 64 steps, minibatch 32, 4 epochs, $\gamma = 0.99$, $\lambda = 0.95$, entropy 0.01, clip 0.2, value coefficient 0.25
Learning rate	3.5×10^{-4}, $\times 0.7$ after an epoch with $> 99\%$ success, floor 10^{-6}
Budget	40 epochs (655 360 steps); 25 epochs in the five-seed 30×30 study; 20 fine-tuning epochs on faulty chips
Seeds	5 at 30×30 (25 epochs), 50×50, 60×60 and on faulty chips; 1 otherwise
Hardware	NVIDIA Tesla T4 and A100 GPUs, AMD EPYC 7742 CPUs

VI. Results

We first validate our implementation of the baseline and then compare the two methods on 30×30 chips, across chip sizes and on chips with faulty MCs.

A. Validation of the Baseline Implementation

To verify our implementation of the baseline, it and the authors' original implementation [8] (commit `1667016`, unmodified) were trained with the settings of the authors' logged 30×30 run for 40 epochs, with three seeds each. The original implementation at this commit uses a smaller CNN (32, 64 and 64 filters, FC 128) and a learning-rate factor of 0.7; our first set of runs used the Table I network and the factor 0.5 of the authors' log, and a second set used the original network and factor. Table II and Fig. 1 show the results. Our implementation reached 95% success earlier on average (Welch's t-test, $p = 0.089$ and 0.039) and ended 0.8 and 1.2 percentage points lower; after Holm correction over the four metrics, only the lower final success of the second set remains significant ($p = 0.028$). The comparison gives no indication that our implementation of the baseline learns more slowly than the original.

TABLE II
Comparison With the Authors' Original Implementation (Three Seeds Each)

Implementation	First epoch ≥95%	Converged (epoch)	Final success (%)	Final cycles
Original [8]	25.7 ± 3.1	25.7 ± 3.1	99.81 ± 0.15	7.95 ± 0.61
Ours, Table I CNN	20.3 ± 0.6	23.3 ± 1.5	99.05 ± 0.26	8.42 ± 0.14
Ours, original CNN	18.7 ± 1.5	20.7 ± 3.5	98.64 ± 0.28	8.70 ± 0.20

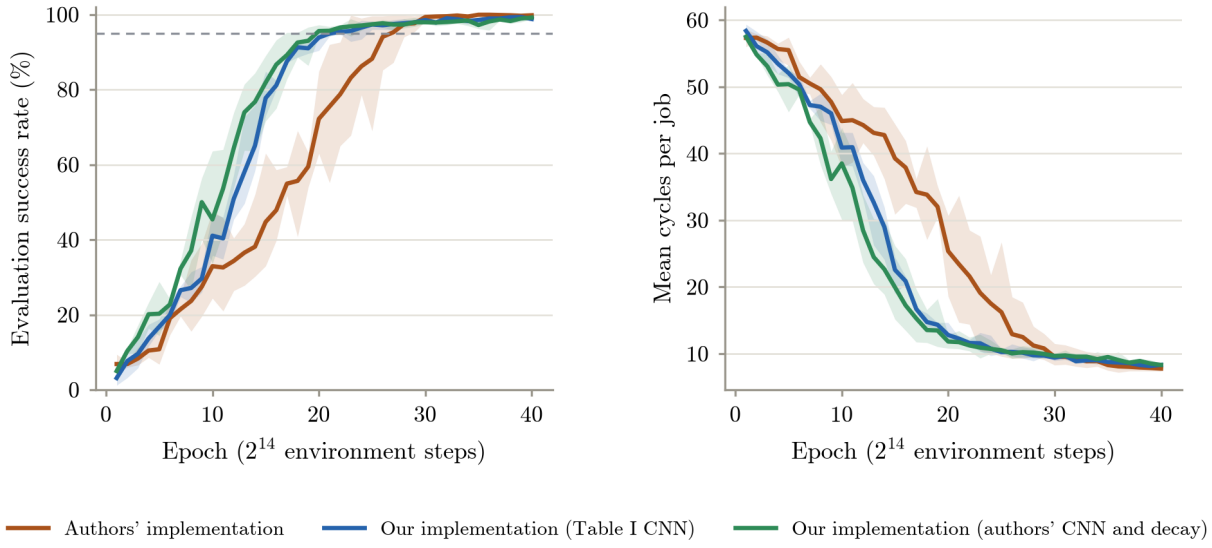


Fig. 1. Per-epoch success rate (left) and mean cycles (right) of the authors' original implementation and of our implementation of the baseline under the settings of the authors' logged run (three seeds each; line: mean, band: minimum to maximum). The dashed line marks 95% success.

B. Comparison With the Results Reported in the Base Paper

Elfar et al. [5] report the training curves of their CNN on healthy chips in Fig. 7 of their paper (five runs, 500 random jobs evaluated after every epoch). Table III compares the values read from that figure with the authors' released implementation under the settings of their logged run (Section VI-A), with our implementation of the CNN under the published settings, and with the direction-aware GCN. The values for [5] are read from a plot and are approximate; the first plotted point is labelled epoch 0 there and corresponds to the first evaluation here.

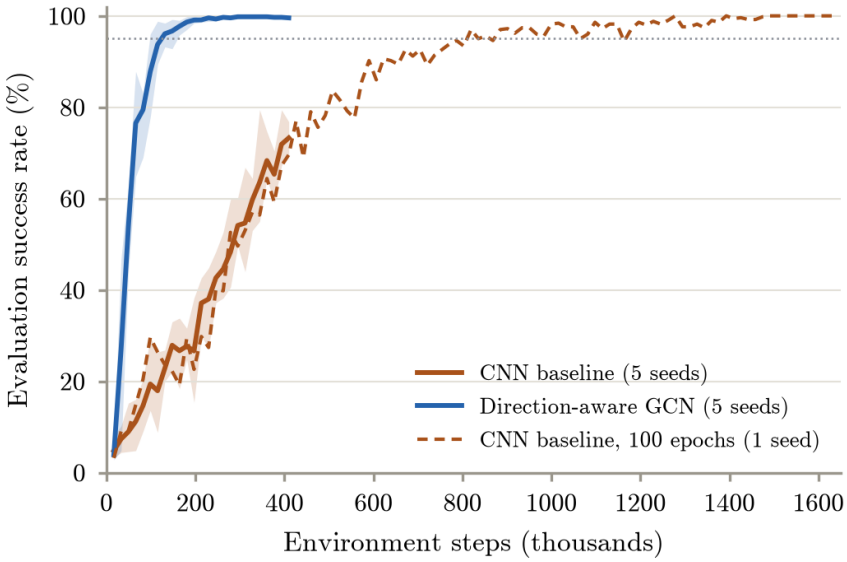


Fig. 2. Per-epoch evaluation success rate on 30×30 chips: five seeds of each method (line: mean, band: minimum to maximum) and a 100-epoch run of the CNN. The dotted line marks the 95% convergence threshold.

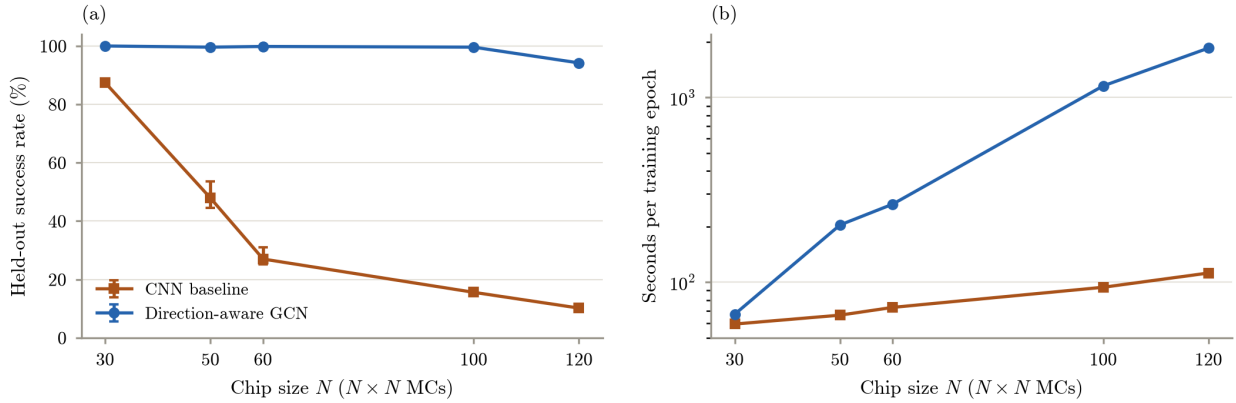


Fig. 3. Chip-size study (40 epochs): (a) held-out success rate (mean and range over five seeds at 50×50 and 60×60, one seed otherwise); (b) mean wall-clock time per epoch (training and 500-job evaluation) on a Tesla T4 GPU, logarithmic scale.

The reported curves start at 50–70% success and reach 100% within 10 to 25 epochs. Neither the authors' released implementation, under the settings of their logged run, nor our implementation of the CNN under the published settings attains this: both start below 10% success, and the CNN needs 23–29 and 50 epochs respectively to converge on 30×30 chips. The direction-aware GCN converges within the 6–10 epochs reported for the CNN on 30×30 chips, but it too starts from a low success rate. The high initial success rates in [5] therefore suggest an easier task distribution or evaluation protocol than the one described there; candidate differences (collision cues in the observation, the episode limit, the droplet sizes and the treatment of timeouts) are the subject of an ongoing ablation.

C. *Learning Speed on* 30×30 *Chips*

Table IV and Fig. 2 compare the two methods over five seeds with a budget of 25 epochs. The direction-aware GCN met the convergence criterion in every seed, between epochs 6 and 10, and routed $99.9 \pm 0.1\%$ of the held-out jobs. The CNN converged in no seed and routed $73.7 \pm 4.0\%$ ($p = 1.2 \times 10^{-4}$); its highest and the GCN's lowest success rates are 21 percentage points apart. When the CNN was trained for 100 epochs with one seed, it met the convergence criterion at epoch 50, after 819 k environment steps, about six times as many as the GCN, and finally

TABLE III
TRAINING SPEED REPORTED IN [5] (FIG. 7) AND MEASURED HERE (HEALTHY CHIPS)

Chip	Agent	Runs	Success at first evaluation (%)	Epoch at which 95% success is held*
30×30	CNN, reported in [5]	5	≈65–70	≈8–10†
	CNN, released code [8], logged settings	3	6.9	23–29
	CNN, our implementation	5	4.8	none within 25; 50 in a 100-epoch run
	Direction-aware GCN	5	3.9	**6–10**
60×60	CNN, reported in [5]	5	≈55–60	≈8–10†
	CNN, our implementation	5	1.2	none within 40 (best 34.8%)
	Direction-aware GCN	5	3.3	**13–21**
120×120	CNN, reported in [5]	5	≈50	≈20–25†
	CNN, our implementation	1	3.2	none within 40 (best 14.2%)
	Direction-aware GCN	1	2.0	none within 40 (best 95.6%)

*Convergence criterion of Section V (first of three consecutive evaluations ≥95%). †Epoch at which the plotted success rate reaches 100%.

routed 100% of the held-out jobs in 9.72 cycles. On 30×30 chips the advantage of the graph encoder is therefore its learning speed and its size, not its final performance.

TABLE IV
FIVE SEEDS ON HEALTHY 30×30 CHIPS (25 EPOCHS)

Method	Success (%)	Mean cycles	Invalid actions per decision	Steps to convergence
CNN	73.7 ± 4.0	18.62 ± 0.92	0.18 ± 0.03	none of 5
DA-GCN	$\mathbf{99.9 \pm 0.1}$	$\mathbf{10.16 \pm 0.41}$	$\mathbf{0.003 \pm 0.003}$	$\mathbf{134 \pm 24}$ k

D. Chip Size

Both methods were trained from scratch with 40 epochs on chips of 30×30 to 120×120 MCs (Table V, Fig. 3). The success rate of the CNN, which observes every chip resampled to 30×30, fell from 87.4% to 10.2%, and it converged at no size. The direction-aware GCN routed on average at least 99.6% of the held-out jobs (every run at least 99.2%) up to 100×100 MCs and converged at each of these sizes; on 120×120 chips it reached 94.2% without meeting the convergence criterion, with per-epoch success between 89.8% and 95.6% over the last five epochs. Over five seeds the CNN routed $47.9 \pm 3.7\%$ against $99.6 \pm 0.3\%$ for the GCN on 50×50 chips and $27.0 \pm 2.5\%$ against $99.8 \pm 0.2\%$ on 60×60 chips ($p = 5.2 \times 10^{-6}$ and 2.7×10^{-7}). For the single-seed sizes, the jobs solved by only one method were almost all solved by the GCN (exact McNemar test, $p < 10^{-18}$ at every size).

Computational cost. The cost of the GCN grows with the number of MCs, whereas the CNN always processes a 30×30 input. On a Tesla T4 an epoch (training and the 500-job evaluation) took 67 s on 30×30 chips and 1 852 s on 120×120 chips for the GCN, against 59 s and 112 s for the CNN (Fig. 3b); per forward pass, the GCN needs more multiply–accumulate operations than the CNN (230 M) from about 55×55 MCs.

E. Chips With Faulty MCs

Following the protocol of [5, Fig. 8], the healthy 30×30 agent of each seed was fine-tuned for 20 epochs on chips with 10% or 20% fully degraded MCs, which appear in the health channel and must be avoided. Table VI reports the held-out results for five seeds. The direction-aware GCN routed 94.0% and 82.2% of the jobs against 80.4% and 69.3% for the CNN ($p = 3.6 \times 10^{-4}$ and 3.8×10^{-6}), in fewer cycles. Neither method met the convergence criterion within the 20 epochs, and the CNN started from healthy agents that had not converged, so part of this gap is inherited from the healthy training.

TABLE V
CHIP-SIZE STUDY (40 EPOCHS; HELD-OUT SUCCESS IN %, MEAN CYCLES IN PARENTHESES)

Chip	Seeds	CNN	DA-GCN
30×30	1	87.4 (14.52)	**100.0 (9.84)**
50×50	5	47.9 ± 3.7 (38.94)	$\mathbf{99.6 \pm 0.3}$ **(17.25)**
60×60	5	27.0 ± 2.5 (53.76)	$\mathbf{99.8 \pm 0.2}$ **(21.03)**
100×100	1	15.6 (86.47)	**99.6 (35.05)**
120×120	1	10.2 (108.61)	**94.2 (47.24)**

DA-GCN steps to convergence: 147 k, 246 ± 60 k, 259 ± 55 k, 246 k and not reached; the CNN converged at no size.

TABLE VI
CHIPS WITH FAULTY MCS (30×30, FIVE SEEDS, 20 FINE-TUNING EPOCHS)

Faulty MCs	Method	Success (%)	Mean cycles	Invalid actions per decision
10%	CNN	80.4 ± 2.9	17.47 ± 0.51	0.102 ± 0.040
	DA-GCN	$\mathbf{94.0 \pm 0.5}$	$\mathbf{13.15 \pm 0.18}$	$\mathbf{0.020 \pm 0.013}$
20%	CNN	69.3 ± 1.9	21.81 ± 0.33	0.087 ± 0.020
	DA-GCN	$\mathbf{82.2 \pm 1.6}$	$\mathbf{17.85 \pm 0.57}$	$\mathbf{0.045 \pm 0.023}$

VII. DISCUSSION AND LIMITATIONS

The results suggest that the inductive bias of the encoder, rather than its capacity, governs how quickly the router learns: the direction-aware GCN applies the same local rule at every position, whereas the fully connected layer of the CNN must learn the mapping for every position separately. The training curves reported for the base method could not be reproduced with its released implementation or with ours (Table III), so our baseline should be compared with the published curves with care. Several limitations qualify these findings. The comparisons at 30×30 (40 epochs), 100×100 and 120×120 use one seed, and runs of the GCN with the same seed on different platforms varied considerably on 120×120 chips (per-epoch success after 21 epochs: 86.4% on the T4, 60.6% on the A100). Above 30×30 MCs the CNN observes a resampled chip while the GCN observes the native grid, and the two encoders also differ in readout and width, so the comparison does not isolate the graph representation from these factors. Each chip size was trained from scratch, unlike the transfer learning of [5]. On healthy chips the sensed health of almost every MC stays at its maximum during an episode, and the faulty chips contain only fully degraded MCs; the gradual wear of MCs over many bioassays is not yet exercised. Finally, within the 20-, 25- and 40-epoch budgets the learning-rate decay was triggered only for the GCN, the only method that exceeded 99% success; in the 100-epoch run of the CNN it was first triggered at epoch 79.

VIII. CONCLUSION

This paper proposed a direction-aware graph representation of the MEDA electrode array for reinforcement-learning-based droplet routing. Direction information is necessary in message passing, since isotropic aggregation cannot distinguish a routing job from its mirror images, and the resulting encoder is a size-independent fully convolutional network with 76 k parameters. Compared with the CNN–PPO method of [5] with the same simulator, reward and PPO settings, it converged in about one sixth of the environment steps on 30×30 chips (five GCN seeds against a single 100-epoch CNN run), maintained on average at least 99.6% success up to 100×100 MCs within a budget at which the CNN fell to 15.6%, and routed more jobs on chips with faulty MCs. Future work will extend the multi-seed evaluation to the largest chips, separate the effects of representation, readout and observation resolution, evaluate transfer across chip sizes without retraining, and study routing under gradual degradation and on bioassay benchmarks.

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